Heterogeneous effects of monetary policy in wartime Russia

Abstract

The present study explores the strength of the transmission channel of monetary policy on sectoral inflation and output in Russia before and after the start of the war in Ukraine in February 2022. The results using panel Local Projections (LP) with ample macroeconomic controls provide evidence for an attenuation of the demand channel and an exacerbation of the output/inflation trade-off after February 2022, partially explained by sanctions, war-related demand, and government ownership shares. Furthermore, firm and household inflation expectations are identified as a relevant factor for explaining inflation dynamics, whose omission helps explain the appearance of a price puzzle in previous Russian empirical studies.

Keywords: *Russian monetary policy, Russia-Ukraine war, monetary policy transmission mechanism, price puzzle, Panel Local Projections*

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Appendix I

1. Introduction

In the early days and weeks of the war in Ukraine it was expected that the severe restrictions on exports to Russia, the massive exodus of Western firms from the country, the isolation of its banking system, and the freezing of the Central Bank's assets would irreparably cripple the Russian economy and force it into a hyperinflationary recession. However, after more than 4 years of bloodshed it is evident that the Russian economy managed to, at least in the short-term, outperform expectations as it avoided a collapse of the ruble, redirected some portion of the trade to Asia, and managed to bring inflation to single digits, albeit still above its target of 4%. While this result can partially be explained by the fact that the sanctions imposed on Russia targeted its technological development and longer-term economic growth and weren't supported by the whole world, both Western and Russian independent media cited the Central Bank of Russia and, especially, its head Elvira Nabiullina as significant contributors to the unusual resilience of the Russian economy.

At the same time, the policy of the Central Bank has seen some intense criticism from within the economic elite in Russia, as tightened credit conditions made existence more difficult even for state-owned firms. It could be argued that some sectors found themselves between the hammer and the anvil, having to compete with the military sector in salaries, while suffering from scarcity or priciness of imports on the one side, and having to pay higher interest rates to finance current inventories and wages on the other. This, in turn, would either force them to cut production, reduce margins, increase prices, or a combination of the three. This study attempts to empirically test whether there is significant heterogeneity in terms of price and output responses to a monetary policy shock between different sectors, which would offer useful insights into the unintended consequences of monetary policy during a high-stress period such as war or international sanctions.

2. Literature review

Monetary policy during wartime has historically held a rather accommodative stance vis-à-vis the government, showing willingness to tolerate inflation and financial repression to facilitate military spending and debt management, as in the UK and US during WWI (Kool, 1995), US during WWII (Hetzel and Leach, 2001), Israel between 1970 and 1985 (Beenstock and Ben-Gad, 1989). In this regard, the Bank of Russia stands in stark contrast to the behaviour of its predecessors, continuing to target price stability even at the expense of the government and the military sector.

Empirically, the analysis of monetary policy, especially during war, is made difficult by the presence of a price puzzle, which has been a recurring issue in the empirical literature since late-20th century (Sims 1992; Eichenbaum, 1992). It arises due to the inherent endogeneity of monetary policy to the current state of the economy and expectations about it, which results in counterintuitive results, wherein contractionary monetary policy leads to an increase in prices, domestic currency depreciation, and other peculiarities. It has therefore become increasingly important to isolate the exogenous variation of policy variables to be able to discern clean effects.

One explanation for this puzzle has been an imperfect control for the information set of the regulator. Sims (1992) suggested that including commodity prices helped mitigate a portion of the price puzzle. Bernanke et al. (2005) introduced the FAVAR approach, which allowed to account

for more than a hundred variables in the regulator's information set by summarising them into several latent factors. Shestakov (2017) applied a FAVAR in the Russian case, which eliminated the price puzzle and found no evidence of a cost channel at the aggregate level.

Another common method for dealing with monetary policy endogeneity is by using monetary surprises. While the current literature tends to use surprises as an external instrument (Gertler and Karadi, 2015; Stock and Watson, 2018), it is usually asymptotically equivalent and often inefficient in comparison to using it directly in a one-step regression (Kuttner, 2001; Lloyd and Manuel, 2024).

In the empirical analysis of emerging economies (EMEs) the price puzzle has been an often-recurring consequence of suboptimal identification, appearing in around half of the studies analysed in a meta-study by Rusnák et al. (2013). In open economies the price puzzle often arose as a result of the foreign exchange (or FX) puzzle, wherein the domestic currency depreciated after a monetary shock, which spilled over to inflation. Ha et al. (2024) have found that both puzzles can often be dealt with by including forward looking variables that measure inflation expectations as well as equity indices that capture market sentiment. In their India study Ghosh et al. (2021) also included Economic Policy Uncertainty as an endogenous variable in the system that helped capture economic sentiment, close in spirit to Romer and Romer's (2004) narrative approach.

In the Russian case, Abramov et al. (2022), due to data scarcity and unlike literature focusing on the US, used daily changes in different segments of the yield curve as surprise measures. This is common practice in cross-country studies covering developing countries (Choi et al., 2022), as well as in Russia (Bannikova, 2024), due to relative illiquidity of financial markets in EMEs. Moreover, thanks to a policy known as the "week of silence" at the Bank of Russia (BoR) whereby policymakers don't speak publicly a week before the key rate decision, the daily change can be plausibly assumed to be exogenous. Lastly, it is worth noticing that Russian papers that do use intraday data on FX futures see no significant improvement in results with the presence of a persistent price puzzle even with the usage of a 30-minute window (Tishin, 2019).

3. Russian monetary and economic policy background

Before turning to the empirics, it's important to summarise the institutional, economic, and monetary policy context in Russia. In September 2013 the BoR introduced the key rate as its main policy tool and later in November 2014 transitioned to a free float regime. Finally, in 2015 the Bank switched to inflation targeting, setting the desirable level at 4%. After the invasion of Ukraine in February 2022, the consequent separation of Russian credit institutions from the global financial system, and the gradual introduction of severe sanctions, the BoR implemented restrictions on FX purchases, bond coupon and dividend payments in foreign currency, as well as cross-border fund transfers to non-residents, all to limit ruble depreciation and prevent capital flight. This also coincided with the introduction of an obligation for exporters to sell a portion (the exact percentage varied over time) of their earnings in foreign currency on the domestic market, as well as with a drastic increase in government spending to finance the war effort.

In 2022, the government budget contribution to the growth in gross fixed capital formation increased significantly, staying at high levels through 2023 (Simola, 2024). Simola (2023) classifies "Fabricated metal products, except machinery and equipment", "Computer, electronic

and optical products", and "Other transport equipment" (which includes drone manufacturing, that skyrocketed after the invasion) as war-related industries and finds that they considerably outperformed other industries in terms of output since the start of the war and significantly contributed to Russia's economic rebound in 2023.

The impact of sanctions on the Russian economy deserves special attention. Through a DiD approach Egorov et al. (2025) find that imports of sanctioned products were down 27% through 2023 with firms exposed to sanctions seeing on average a 17% slump in their revenues in the first 2 war years. While the effect of sanctions was particularly pronounced in the science and technology sector, which curtailed Russia's long-term growth prospects, manufacturing was also significantly affected. Furmanov and Turovets (2024) identify "Motor vehicles, trailers and semi-trailers", "Wood and products of wood and cork, except furniture", and "Fabricated metal products, except machinery and equipment" as the most impacted by sanctions.

In terms of inflation, the initial slump of imports and the depreciation of the ruble as a result of Western sanctions spurred price growth in 2022 by 11.7 p.p. (Benchimol and Palumbo, 2024). Later, in 2024 and 2025, once sanctions started affecting exporting industries and depressing overall economic activity the impact of sanctions can be considered ambiguous, albeit there are no formal studies to the author's knowledge addressing this specific question.

Another relevant source of inflationary pressure since the start of the war has been the labour market, which since the partial mobilisation in September 2022 and the anti-migration policy turn since March 2024 has created a severe labour shortage in the Russian economy. In March 2026 51% of firms participating in the BoR's survey reported having worker shortage, which shows a decreasing trend since maximas in 2025, but is still well above pre-war levels (Bank of Russia, 2026).

Lastly, an important characteristic of the Russian economy is the state dominance in the banking sector. According to the latest available formal estimates, state-owned or quasi-state-owned institutions accounted for 70% of all bank assets, making it the top sector in terms of government participation (Analytical Credit Rating Agency, 2018). Therefore, heterogeneous effects of monetary policy are of special interest in Russia, as the state domination of its banking system may have an impact on the effectiveness of the conventional demand channel of monetary policy. For instance, Borzykh (2016) finds that a positive shock in the key rate only has a significant impact on actual lending rates for large non-government banks, while state-owned and smaller entities find themselves relatively shielded in this regard.

4. Data

To measure monetary policy surprises, the study uses the changes of the ROISfix rates around the monetary policy meeting at monthly frequency (with maturities from 1 week to 1 year), which are Overnight Interest Swaps (OIS), where the moving leg is based on RUONIA. The 3-month rate has been chosen as a middle ground between short- and longer-term expectations about monetary policy. It's worth noting that results are largely robust to the choice of OIS duration.

These surprise measures have several caveats worth mentioning. First, the ROISfix is published daily at 12:30 Moscow time, which is one hour before the usual meeting time of the BoR.

Therefore, the surprise measure is the difference between the next day's open and the meeting's day open.

There are, however, some exceptions from this rule. The extraordinary meetings on March 3rd 2014, February 28th 2022, April 8th 2022, May 26th 2022, and August 15th 2023 took place before the usual time (13:30) and before the ROISfix publication time, so the difference between the open on the day of the decision and the open of the previous day is treated as the relevant measure. As for the extraordinary December 2014 session, which took place at night between December 15th and 16th, the difference between the 16th open and the 15th open is considered as the surprise measure.

Furthermore, the key rate effectively enters into force only the next day after the policy meeting, which creates issues when the meeting takes place at the end of a month. Therefore, for example, if the rate increase is recorded in November at the BoR, the surprise measure is transferred to that month to preserve its link with its respective monetary decision.

Both aggregate and sector-specific monthly macroeconomic data is gathered from Rosstat (unemployment rate, nominal wages, CPI, PPI, industrial production index), the BoR (USDRUB exchange rate, household and firm inflation expectations), the IMF (commodity terms of trade index developed by Gruss and Kebhaj, 2019).

The main dependent variables are the monthly sector-specific Producer Price (PPI) and Industrial Production (IPI) indices, as well as the ruble exchange rate for the aggregate step. The aggregate part of the study focuses on the period starting in January 2015 up to October 2025 (130 observations), whereas the panel section starts later, in January 2017, when the latest sectoral classification (OKVED2) was introduced (time-series of 106 with a cross section of 37). Finally, sectoral impact of sanctions measures have been retrieved from Furmanov and Turovets (2024).

5. Empirical methodology

To study the effects of monetary policy, Vector Autoregressions (VARs) and Local Projections (LP) are the two most common methods applied in literature. The two approaches find themselves on opposite ends of the bias/variance trade-off: VARs are more efficient, but when incorrectly specified lead to errors compounding over horizons, whereas LP's sacrifice efficiency, while avoiding significant bias by directly estimating separate equations for each horizon (Ramey, 2016). Moreover, LP allow for a more straightforward estimation of state-dependent models by simply including a dummy, rendering it more appropriate for the selected research question.

In this study three LP models are estimated. First, an LP regression is performed at the macroeconomic level to test the reaction of aggregate production and consumer prices, industrial production, and the exchange rate to monetary policy shocks:

(1) $\Delta log\ p_{t+h} = \alpha_h + \beta_h M_t + \theta_h (D_{war} \times M_t) + \gamma_h \Delta X_{t-1} + u^h_{t+h}$

Where p_{t+h} is the outcome variable of interest (PPI, CPI, IPI, exchange rate) at horizon h and M_{t-1} is the monetary policy shock (3-month ROISFIX rate at the baseline). D_{war} is the war dummy, which takes the value of 1 starting March 2022, as it's hypothesised that responses to a monetary policy shock have changed since the start of the war in Ukraine. The set of controls X_t

includes two lagged first differences of the PPI, industrial production index, the dollar exchange rate, export commodity prices, the stock-market index, and aggregate unemployment rates, as well as lagged surprise measures. Moreover, firm inflation expectations are included both lagged and contemporaneously, as their omission has been found to explain a large portion of the price puzzle in developing countries (Ha et al., 2024). The inclusion of contemporaneous inflation expectations could be considered a limitation of the present study as it is unclear whether the Bank of Russia has access to the same month's data at the time of the rate decision, since it's often officially published post-meeting. Nevertheless, this approach has some traction in the literature for EMEs (Ha et al., 2024; Ghosh et al., 2021) and it's not implausible to claim that the data is available to the BoR, the organisation responsible for collecting it, before the publication date.

Then, in order to identify the heterogeneous effects of monetary policy on different sectors, panel local projections are estimated at monthly frequency, interacting the explanatory variable with the sector fixed effects as in Aruoba and Drechsel (2024), which returns sector-specific coefficients. Relatively broad standard errors are to be expected, as estimation at the sectoral level is complicated by the prominence of non-macro components in price fluctuations, which have been found to account for 85% of variation in US data (Boivin et al., 2009) and 62% in Russia (Deryugina et al., 2019).

$$(2)\ \Delta log\ p_{i,t+h} = \alpha_{i,h} + month_t + \beta_{i,h} M_t + \theta_{i,h}(D_{war} \times M_t) + \gamma_h \Delta X_{i,t-1} + u^h_{i,t+h}$$

With long differences of log sectoral PPI and IPI as the dependent variables; lagged first differences of the sector PPI, industrial production index, the dollar exchange rate, export commodity prices, aggregate unemployment ratesas controls; firm inflation expectations enter contemporaneously and with two lags; as well as sector ($\alpha_{i,h}$) and month-of-year ($month_t$) fixed effects to account for seasonality. Standard error clustering is performed at the time level, following suggestions by Almuzara and Sancibrián (2024).

Lastly, an additional analysis to assess the potential reasons behind sectoral heterogeneity in responses is performed, wherein the interaction with the sector fixed effect in Eq. 2 is substituted with either a broader sectoral category (e.g. durables vs non-durables) or a continuous variable (e.g. exposure to sanctions), which is denominated as K_i.

$$(3)\ \Delta log\ p_{i,t+h} = \alpha_{i,h} + month_t + \beta_h(K_i \times M_t) + \theta_h(K_i \times D_{war} \times M_t) + \gamma_h \Delta X_{i,t-1} + u^h_{i,t+h}$$

All regressions are estimated at monthly frequency in long differences in the outcome variable as suggested by Jordà and Taylor (2024), as it is found to be less prone to small-sample bias than estimation in levels, while being asymptotically equivalent. Growth controls enter with two lags. Firm inflation expectations enter contemporaneously and with two lags in their observed units.

Correctly isolating the exogenous monetary shock is another key part of the present paper. Unlike other studies, however, the present analysis neither residualises monetary surprises on the relevant aggregate macroeconomic and financial data preceding the rate decision (Bauer and Swanson, 2022), nor uses it as an instrument for the actual key interest change (Gertler and Karadi, 2015). Instead, the regressions are performed directly in one step with the 3-month ROISFIX rate as the explanatory variable, which, while often equivalent in point estimates to the multi-step "shocks-first" approach as argued by Lloyd and Manuel (2024), is preferred from an inference standpoint

when the controls at the first and second stages differ (as would be the present case, since monetary shocks are inherently aggregate, while responses are sectoral).

A brief analysis showed that the monetary surprises measures were not autocorrelated nor significantly explained by observed macroeconomic variables, so the concern about shock persistence and monetary policy endogeneity to observables can be dropped. Moreover, an aggregate regression with both the one-step and two-step procedures found very similar results in terms of point estimates, with the one-step approach proving itself to be more efficient in comparison. These results are considered complementary and aren't reported.

6. Main results

Figures 1 to 4 show the impulse response functions (IRFs) of aggregate CPI, PPI, IPI, and the exchange rate up to the 24-month horizon. While with a lag of around 12 months, consumer prices fall significantly after a monetary policy tightening. The response of industrial production and producer prices is noisier, but still shows a palpable drop, similar in shape to the findings by Brandão-Marques et al. (2020) for EMEs. It is worth noting that the producer prices are, on the one hand, much noisier due to their more frequent updating and, on the other, much more sensitive to monetary shocks than consumer prices, as evidenced by the magnitude of the responses. The exchange rate of the dollar vis-à-vis the ruble, in contrast, doesn't seem to show a consistently negative response in the long-run, appreciating quite significantly after a brief drop right after the shock. Although this may represent some endogeneity remaining in the monetary shocks measure used, as exchange rate expectations aren't controlled for, it can also be a genuine response of the FX rate on longer horizons. It could be caused by an increase in the risk premium after a contractionary shock, which would be followed by an outflow of capital from government debt. The responsiveness of the exchange rate to monetary policy can be further diluted by the temporary implementation of capital restrictions in early 2022 and the obligation of exporters to sell a certain percentage of their income in foreign currency in the domestic market to keep the ruble strong.

The introduction of a war-dummy doesn't significantly change responses at the aggregate level, slightly attenuating the reaction of the exchange rate and producer prices in the short term. IRFs are reported in the Appendix (Figures A.1 – A.4).

Figure 1: aggregate CPI IRFs to a 1 pp shock of the 3-month ROISfix

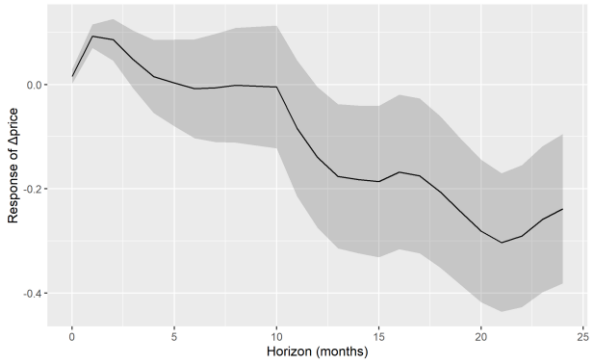


Figure 2: aggregate PPI IRFs to a 1 pp shock of the 3-month ROISfix

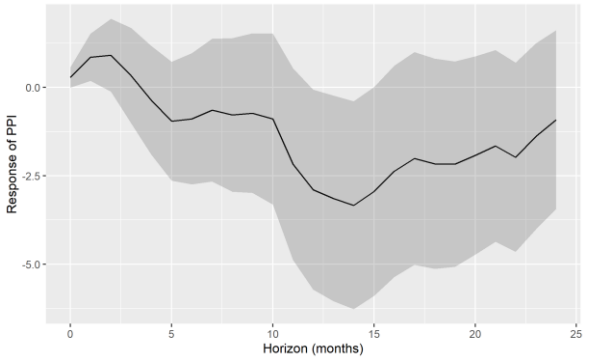


Figure 3: aggregate IPI IRFs to a 1 pp shock of the 3-month ROISfix

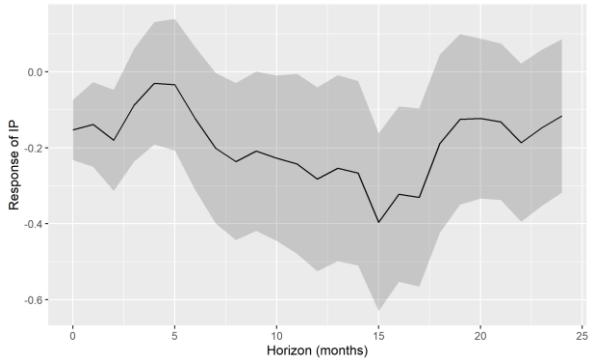
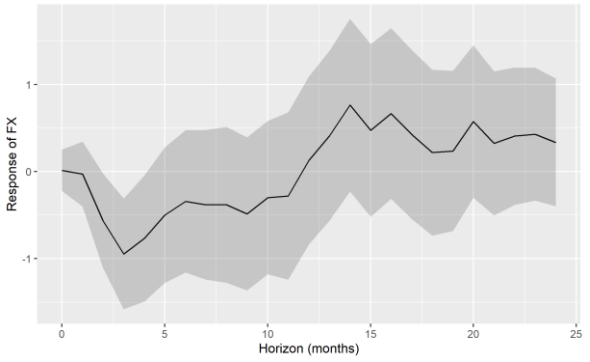


Figure 4: USDRUB exchange rate IRFs to a 1 pp shock of the 3-month ROISfix



Note: all confidence intervals are at 90%, ordinary OLS standard errors are used. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. N = 130

Source: own elaboration

Now, having looked at the aggregate impact of monetary policy in Russia we turn to the sector-level responses (Equation 2), which are reported in Figures 5 and 6. While prices react slowly for the first year or so, they are below 0 for most sectors at the 2-year horizon, especially for the furniture, refined petroleum products, and basic pharmaceuticals sectors. In terms of industrial production, at long horizons the effect is also largely negative in accordance with theory, being strongest for producers of motor vehicles, fabricated metals, as well as "other machinery and equipment" (which includes the production of industrial turbines, hydraulic presses, ovens, among others). Some sectors show positive, albeit statistically negligible, positive responses of production, which, however, shouldn't be interpreted as an output inducing effect of monetary policy. Instead, this finding points to significant sector-specific heterogeneity that can dominate the impact of monetary policy. In summary, despite the wide confidence intervals these preliminary results seem to be fairly consistent with a demand channel of monetary policy transmission, as consumption of durables is more affected by monetary policy than non-durables (Choi et al., 2022).

This last notion is further confirmed by separating sectors into broad categories (durables / non-durables / raw materials / utilities), which yielded more negative price responses (Figure 7) for the raw materials producers and the durables at long-horizons, whereas the response for nondurables

and utilities tended to be smaller, albeit the difference is statistically insignificant. This result is consistent with a classic demand channel transmission, since the consumption of durables (and, hence, the materials used for their production) is depressed more strongly than that of non-durables and utilities, whose consumption patterns are more inelastic.

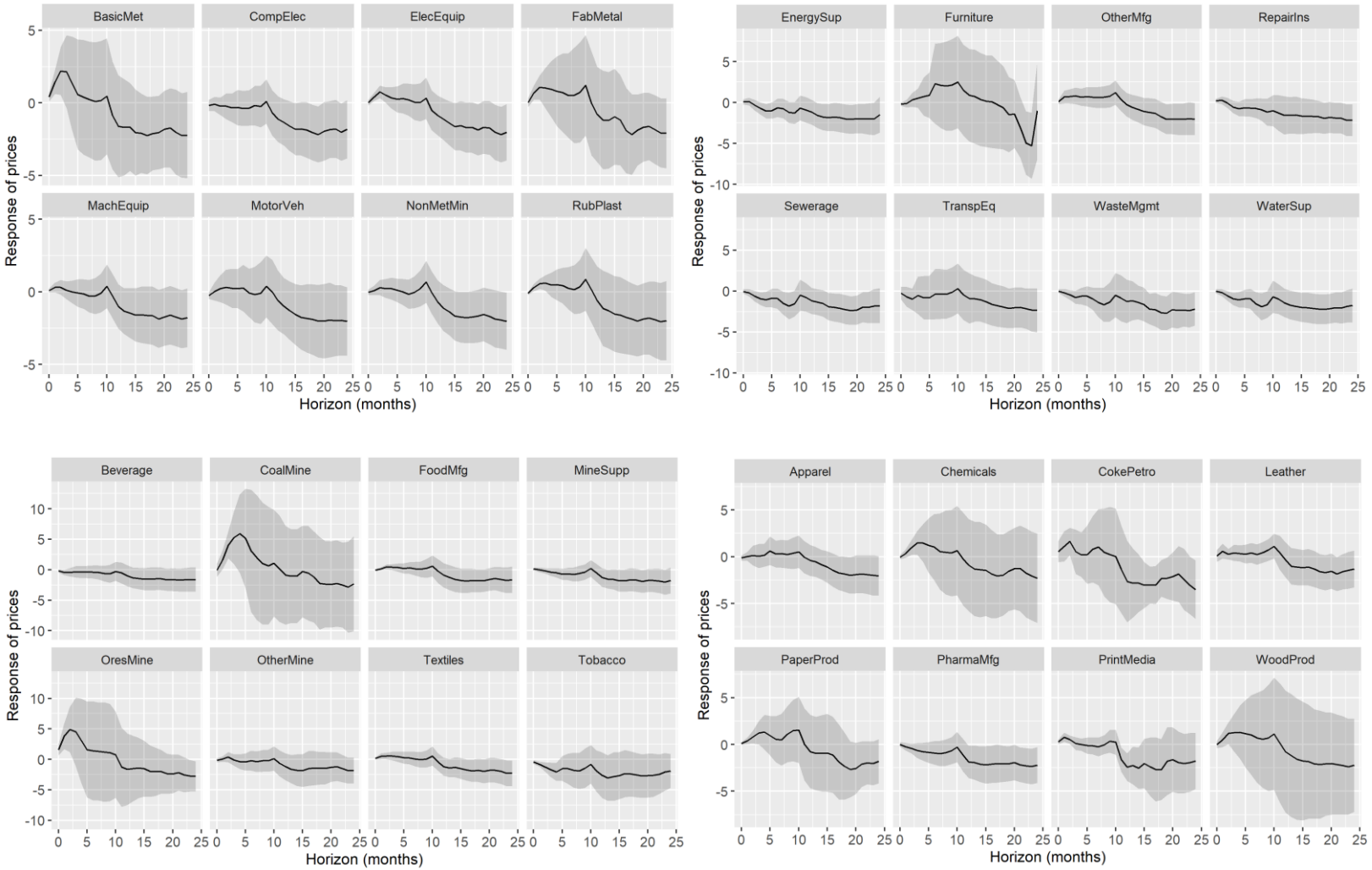
As for sectoral industrial production (Figure 8), the regression using interactions with broader sectors evidences a sharp decline in output among producers of durables, which is significantly larger in magnitude than the responses of utilities, non-durables or raw materials. Furthermore, in line with theory, it is found that the depreciation of the ruble and weakness in export commodity prices also contribute negatively to industrial production.

Now, the same sector-level regressions are performed with the war dummy (Figures 9 and 10). Some sectors see an attenuation of their negative response in prices after the war started, some quite significantly, making the response positive for a prolonged spell of time. Mining (coal and ores), basic and fabricated metal products, furniture, wood, paper, and chemicals manufacturing all show a persistent price puzzle. On the other hand, in terms of production indices, many sectors show a significant exacerbation of the production slump caused by an increase in interest rates, especially computer, electronic and optical products, leather products, and motor vehicles, trailers and semi-trailers. Nevertheless, wood and cork products and basic pharmaceutical products saw, instead, a significant attenuation of the negative effect of a monetary shock after the war started, making output grow in response to a monetary policy tightening.

In terms of broad sectors (Figures 11 and 12), durables and raw materials saw an especially strong attenuation in price responses to a monetary shock, consistent with rising global commodity prices and a heavier than usual production slump in durables. The response of prices on non-durables wasn't weakened as much, despite an exacerbation of the production slump.

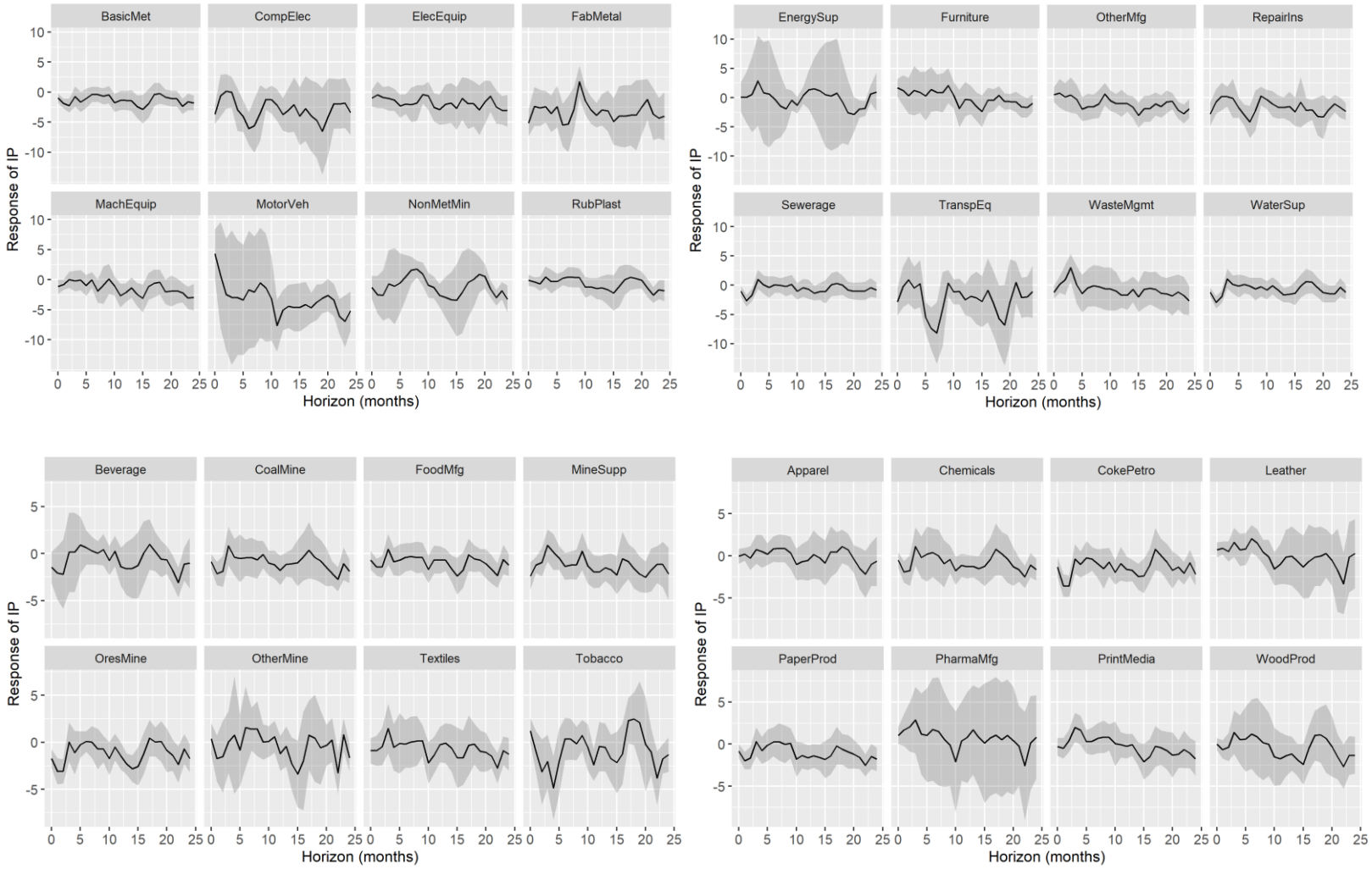
It must be noted that the coefficients interacted with the war dummy must be taken carefully, as the sample size for its estimation is relatively narrow (March 2022 – October 2025, 44 observations), which introduces noise into the measures. Moreover, the analysis yields some counterintuitive results, such as an increase in industrial production in the motor vehicles, pharmaceutical, and energy supply sectors after the start of the war, which hints towards remaining sector-specific components dominating the macro factors in certain industries. The following section attempts to explain these disparities qualitatively and empirically, but further study on sector-specific factors behind inflation and production dynamics in wartime Russia with both a larger cross-section and a longer time-series is required to give a more definite and causal answer.

Figure 5: sectoral PPI IRFs to a 1 pp shock of the 3-month ROISfix



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. N=3922. Source: own elaboration

Figure 6: sectoral IPI IRFs to a 1 pp shock of the 3-month ROISfix



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. N=3922. Source: own elaboration

Figure 7: PPI IRFs to a 1 pp shock of the 3-month ROISfix by broad sector

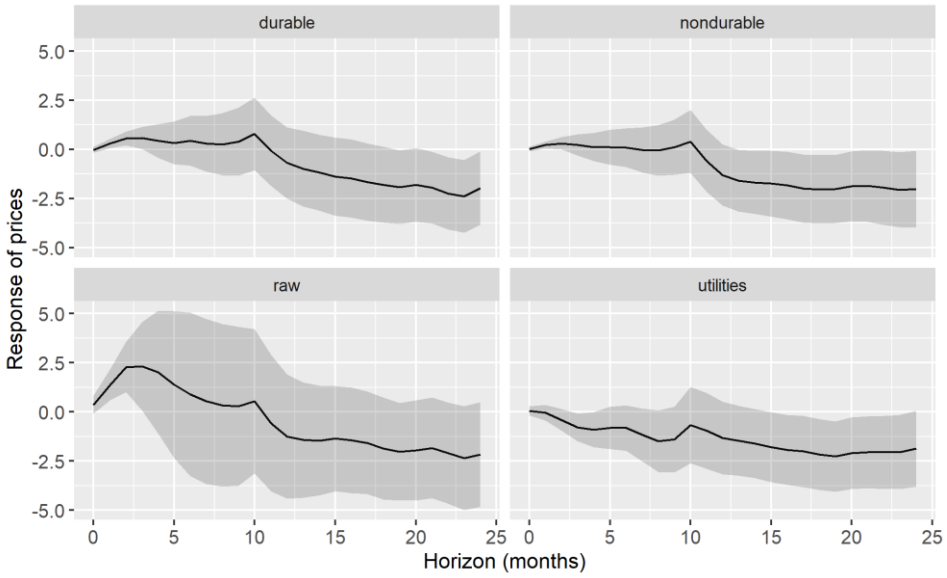
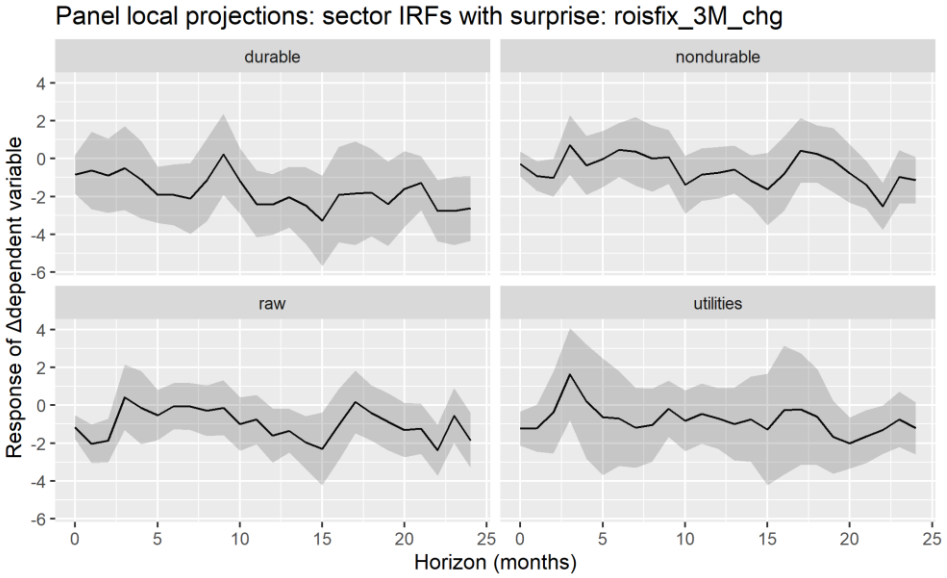
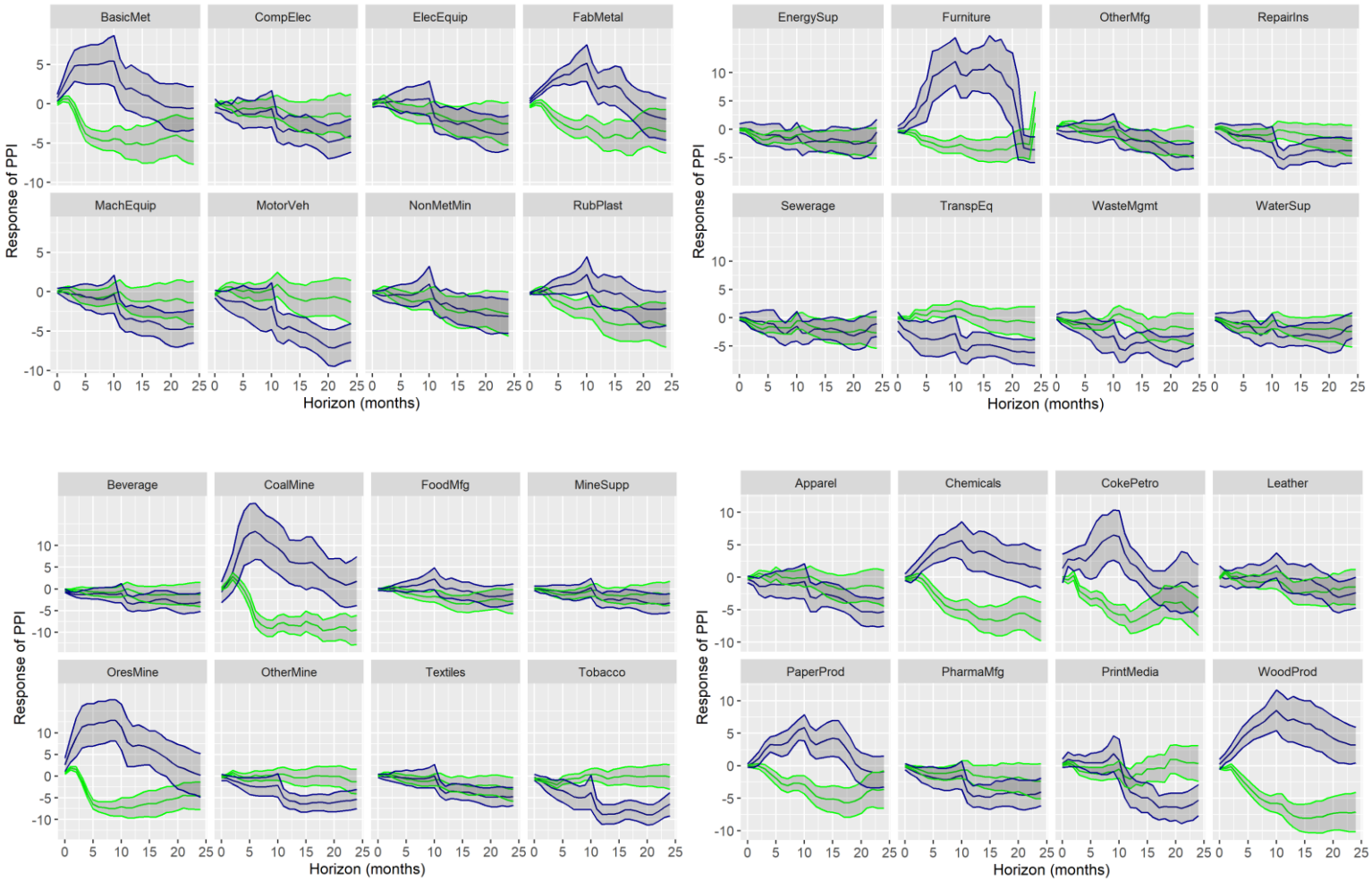


Figure 8: IPI IRFs to a 1 pp shock of the 3-month ROISfix by broad sector



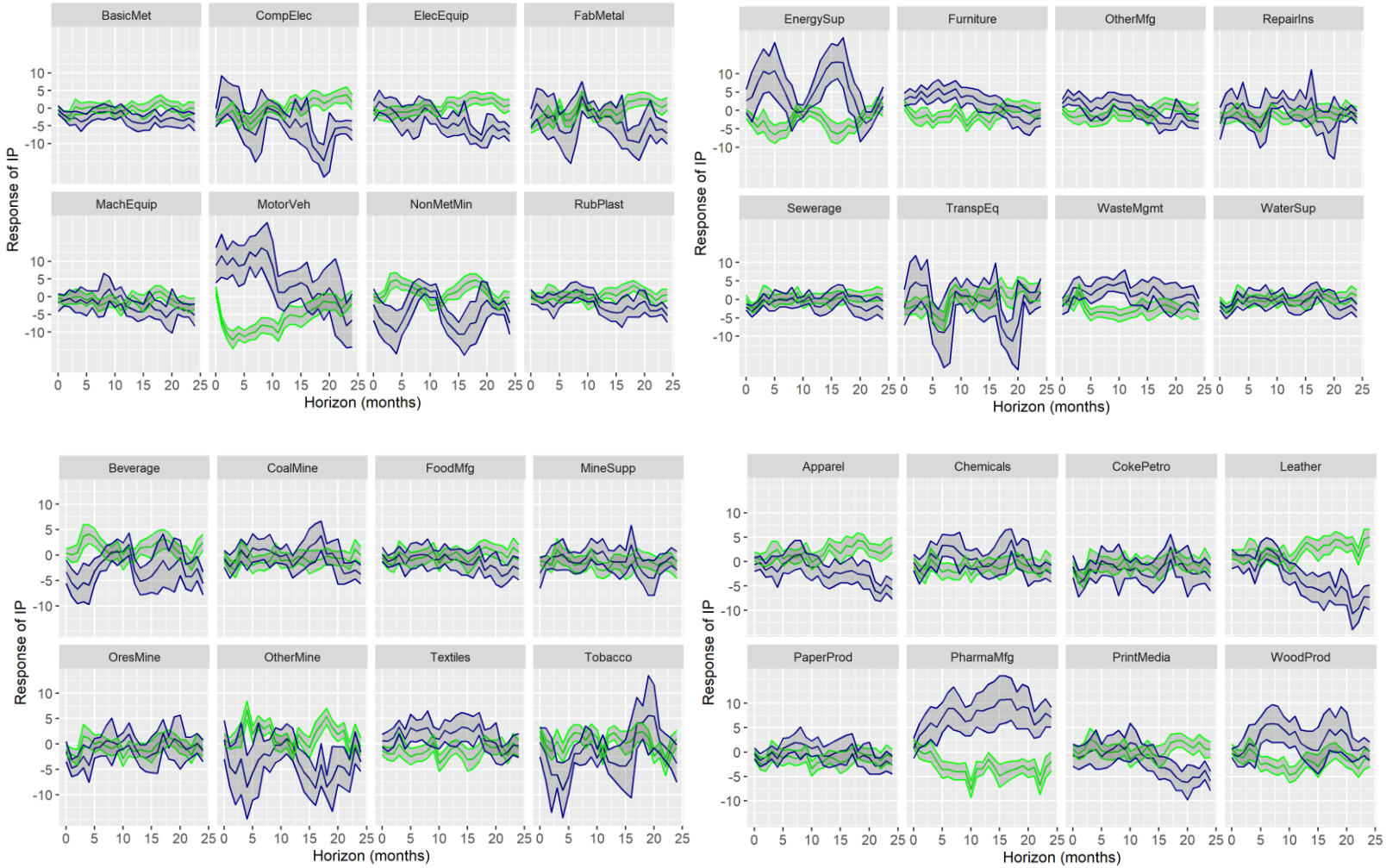
Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. N=3922.
Source: own elaboration.

Figure 9: sectoral PPI IRFs to a 1 pp shock of the 3-month ROISfix before and after February 2022



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. N=3922. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines – before. Source: own elaboration

Figure 10: sectoral IPI IRFs to a 1 pp shock of the 3-month ROISfix before and after February 2022



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines – before. N=3922. Source: own elaboration

Figure 11: PPI IRFs to a 1 pp shock of the 3-month ROISfix by broad sector, before and after February 2022

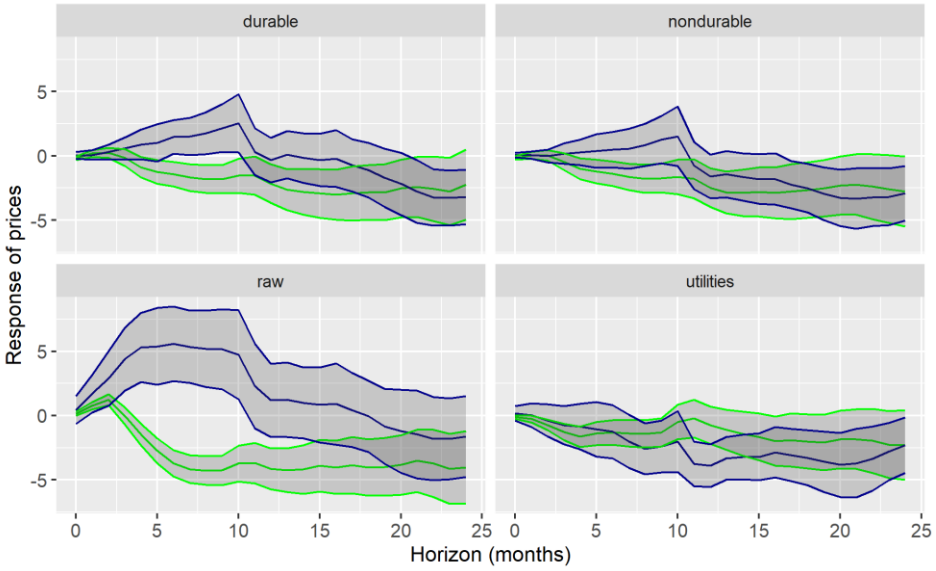
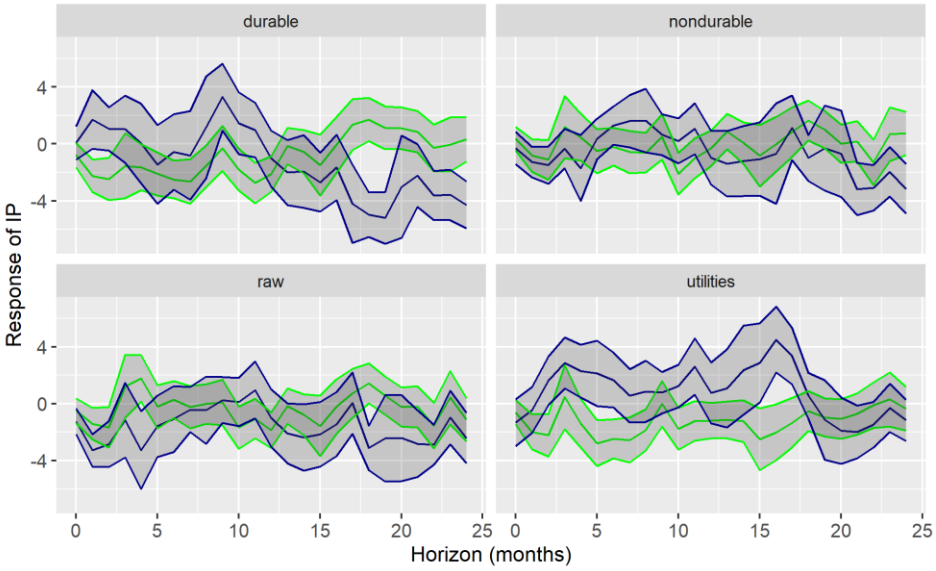


Figure 12: IPI IRFs to a 1 pp shock of the 3-month ROISfix by broad sector, before and after February 2022



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines – before. N=3922. Source: own elaboration.

7. Sectoral heterogeneity before and after February 2022

Here several potential explanations for the change in price responses to monetary policy after the start of the war in Ukraine are proposed and empirically tested with the available data by including the respective variables (impact of sanctions, war-related dummy, government ownership share, working capital dependency) as interactions with the explanatory variable and the war dummy.

Sanctions: It's hypothesised sanctions exert an upward pressure on firm costs, thus limiting the impact of monetary policy on prices and broadening the inflation-output trade-off. To account for a sector-biassed effect of sanctions introduced after the launch of the war in Ukraine estimates of the percentage impact on the output of sanctions on 24 manufacturing industries are drawn from Furmanov and Turovets (2024) and included as interactions with monetary shocks. Excluding two extreme outliers (motor vehicles and wood products manufacturing), whose reaction to monetary policy shocks seems to be dominated by idiosyncratic components, results show that sectors more affected by sanctions suffered a stronger slump in production at all horizons as a result of monetary tightening after the start of the war. Prices, on the other hand, reacted heterogeneously, with no clear exacerbation/attenuation effect from sanctions. Figures A.5 and A.6 show the IRFs by impact group.

Military demand: Military demand is expected to have the opposite effect by lowering exposure to market credit. War-related industries, as defined by Simola (2023), while less reactive to monetary policy shocks before the war, did reduce prices more than industries not directly related to the military machine after February 2022. The behaviour of industrial production for war-related sectors has been more heterogeneous and only started showing significantly negative responses after 16 months, but the reaction was quite massive (-11 p.p. reduction in industrial production per 1 p.p. interest rate increase), which could mean that military demand is incapable of offsetting the reduction in civil demand of durables. IRFs are presented in Figures A.7 and A.8 in the Appendix.

The counterintuitive IPI responses in the motor vehicles manufacturing sector, one could argue, could also be explained by increased government demand for dual-use transportation vehicles (like trucks) for military purposes, which led to the sector's rebound in 2023 and 2024 after a debacle in 2022 as Western producers started leaving the Russian market. Nevertheless, such an explanation would mean that military demand has a non-linear and heterogeneous effect on different sectors, which would require using a more granular sectoral price dataset to be able to quantitatively isolate the mediation effect of war-related demand.

Government ownership: As with military demand, higher government ownership shares can lower firm exposure to market credit. Results show a sizable attenuation effect of government ownership on both price and production responses before the war, but not after (Figures A.9 – A.10 in the Appendix). It is worth pointing out, however, that at least part of the attenuation effect can be attributed to high federal and municipal state ownership in the utilities sector, which is heavily regulated and often subsidised in Russia, reducing the responsiveness of prices to monetary policy. Moreover, like in the case of war-related sectors, industrial production in sectors with higher government ownership shows highly volatile responses to a contractionary monetary shock, exemplified by the bimodal-looking and positive IRF of energy supply shown in Figure 8. Overall, after the start of the war in Ukraine, government ownership does not appear to have insulated producer prices from the contractionary demand channel, being, instead, more relevant for

explaining distortions in the production response, potentially through regulation, subsidies, and public procurement or state orders.

Mobilisation, migration, and labour market: Labour market tightness is expected to enter through an upward push to firm costs. To assess potential confounding emanating from contemporaneous labour market dynamics the nominal wage by OKVED sector, retrieved from Rosstat at monthly frequency, is added as a control. This empirical exercise, however, shows no significant impact of sectoral wage dynamics on the responses of neither prices nor industrial production to monetary policy tightening, which indicates that the lags of the national unemployment level included in the baseline regression seem to largely capture the war-related labour-market distortions.

8. Discussion and conclusion

The results suggest that monetary policy transmission in Russia has weakened since the start of the war, with evidence of an attenuated inflation response and a sharper output–inflation trade-off. The evidence also does not support labour-market distortions or war-related military demand as the main drivers of the observed heterogeneity. Sectoral wage dynamics do not significantly alter the responses of prices or industrial production, while responses of war-related industries remain broadly consistent with a demand-channel interpretation. Sanctions, by contrast, appear to amplify the contraction in industrial production in affected sectors, although their effect on prices is less clear. Government participation attenuated monetary transmission before the war, but this effect becomes less pronounced after 2022.

Overall, the conventional demand channel continues to operate, albeit less cleanly than before the war. Durable-goods sectors experience stronger output declines and greater price moderation than non-durables and raw materials, which is consistent with standard monetary transmission. At the same time, the remaining sectoral heterogeneity points to the importance of unobserved industry-specific factors. The relatively short post-war sample further limits causal interpretation, particularly in sectors with anomalous responses such as energy supply, motor vehicles, and pharmaceuticals.

The results also indicate that the exchange-rate channel has become less reliable. The ruble does not respond consistently to monetary tightening over medium and longer horizons, likely reflecting the role of capital controls, foreign-exchange restrictions, and mandatory export-revenue sales. This weakens BoR's ability to influence domestic prices through the external channel.

Future research could examine sector-specific inflation dynamics in greater detail, separating macroeconomic from microeconomic sources of price variation. Further work should also assess how unconventional monetary measures, capital controls, and financial restrictions have affected the BoR's ability to transmit policy shocks to prices and output.

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Appendix

Figure A. 1: aggregate CPI IRFs to a 1 pp shock of the 3-month ROISfix

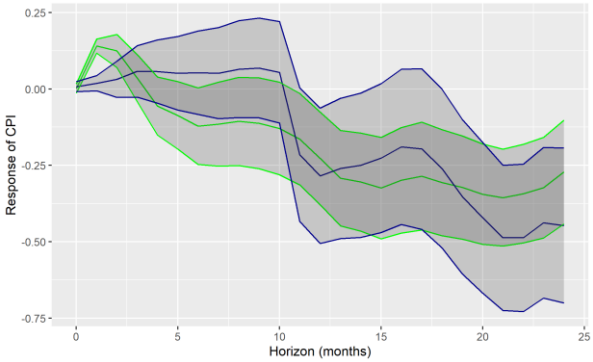


Figure A. 2: aggregate PPI IRFs to a 1 pp shock of the 3-month ROISfix

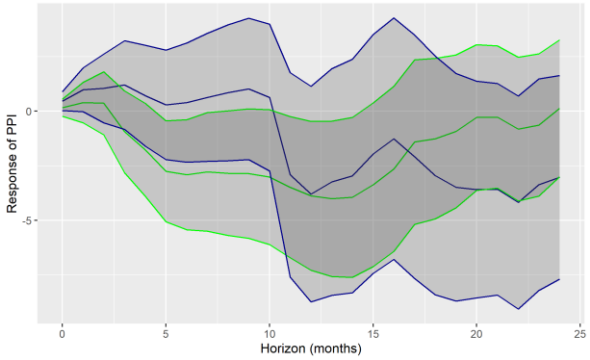


Figure A. 3: aggregate IPI IRFs to a 1 pp shock of the 3-month ROISfix

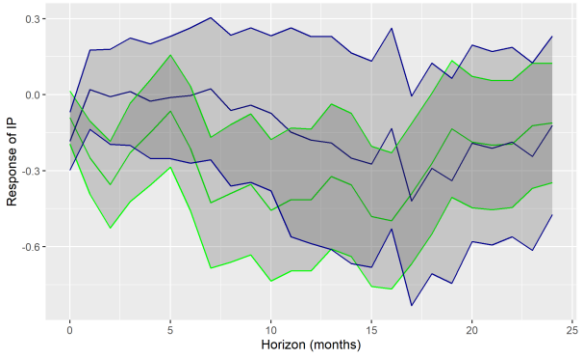
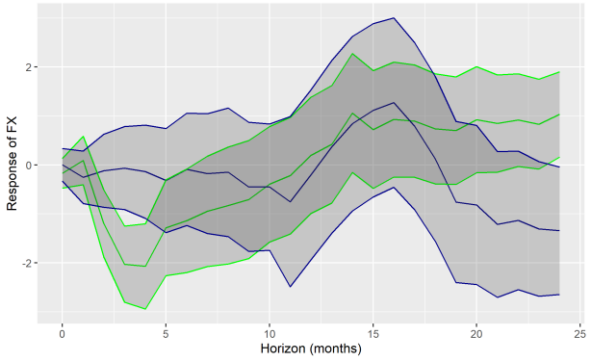


Figure A. 4: USDRUB exchange rate IRFs to a 1 pp shock of the 3-month ROISfix



Note: all confidence intervals are at 90%, ordinary OLS standard errors are used. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines - before. N=130. Source: own elaboration

Figure A. 5: sectoral PPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, grouped by impact of sanctions

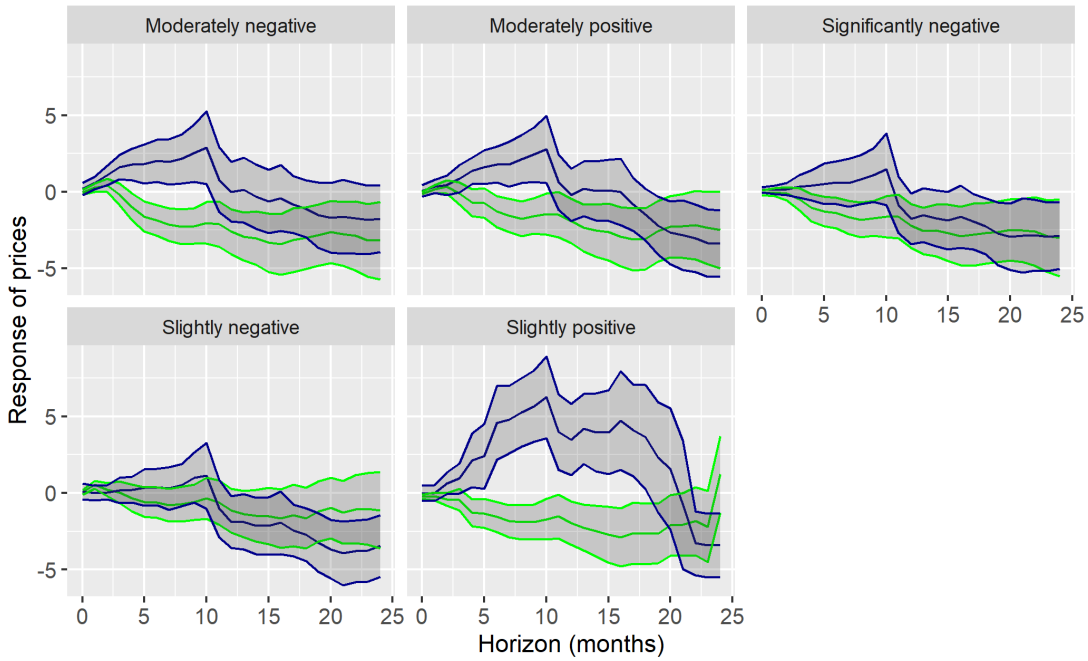
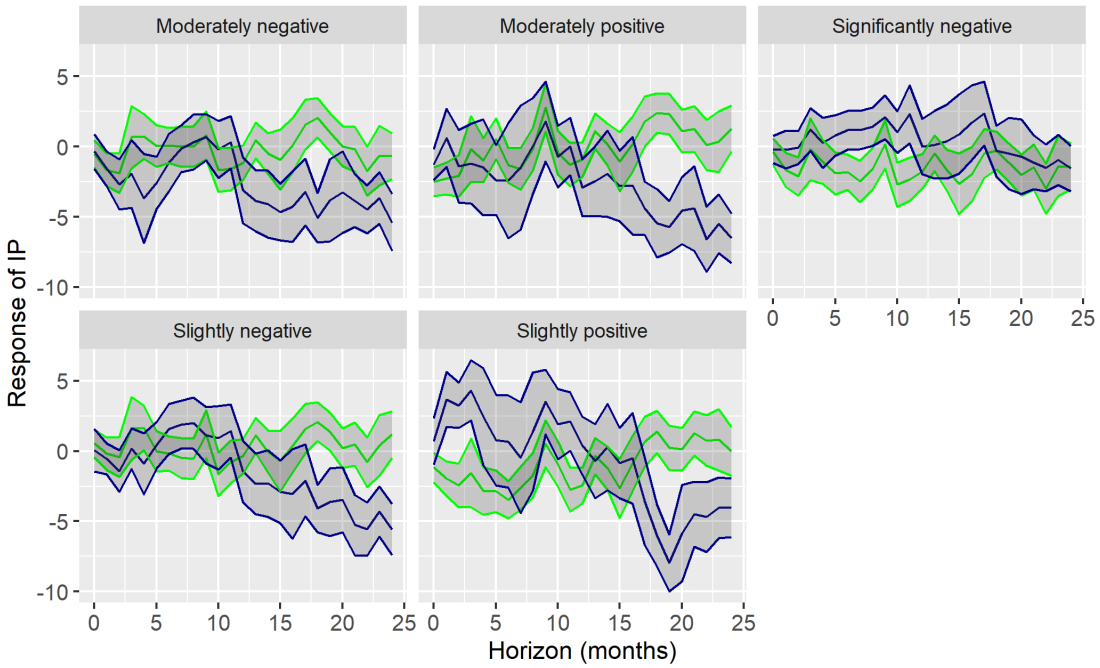


Figure A. 6: sectoral IPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, grouped by impact of sanctions



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines - before. N=3922. Source: own elaboration

Figure A. 7: sectoral PPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, war-related vs non-war-related

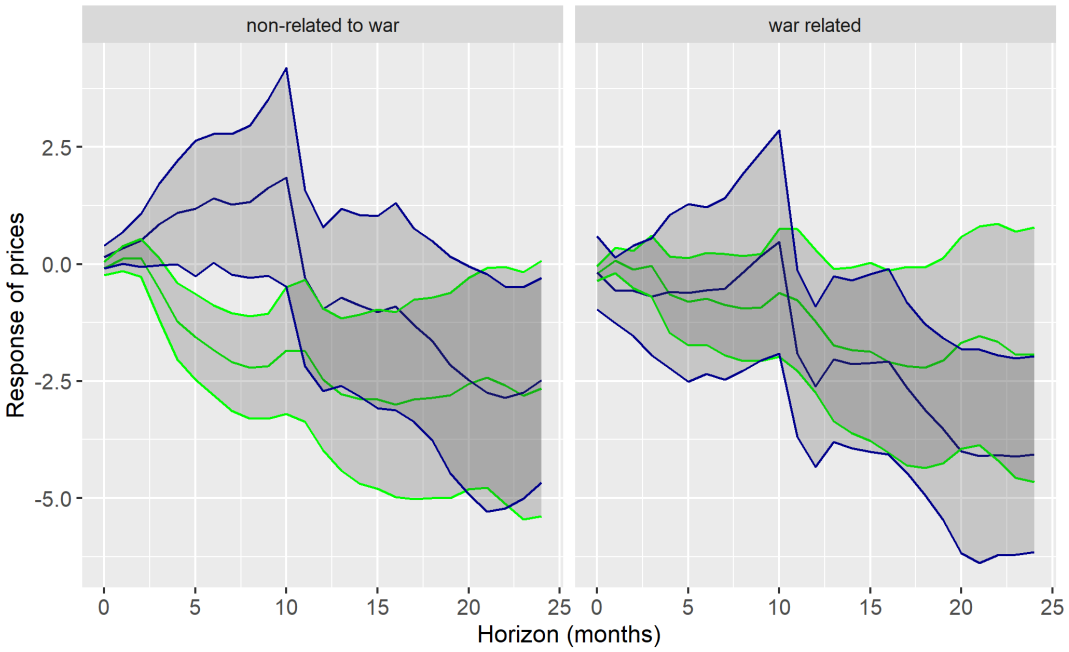
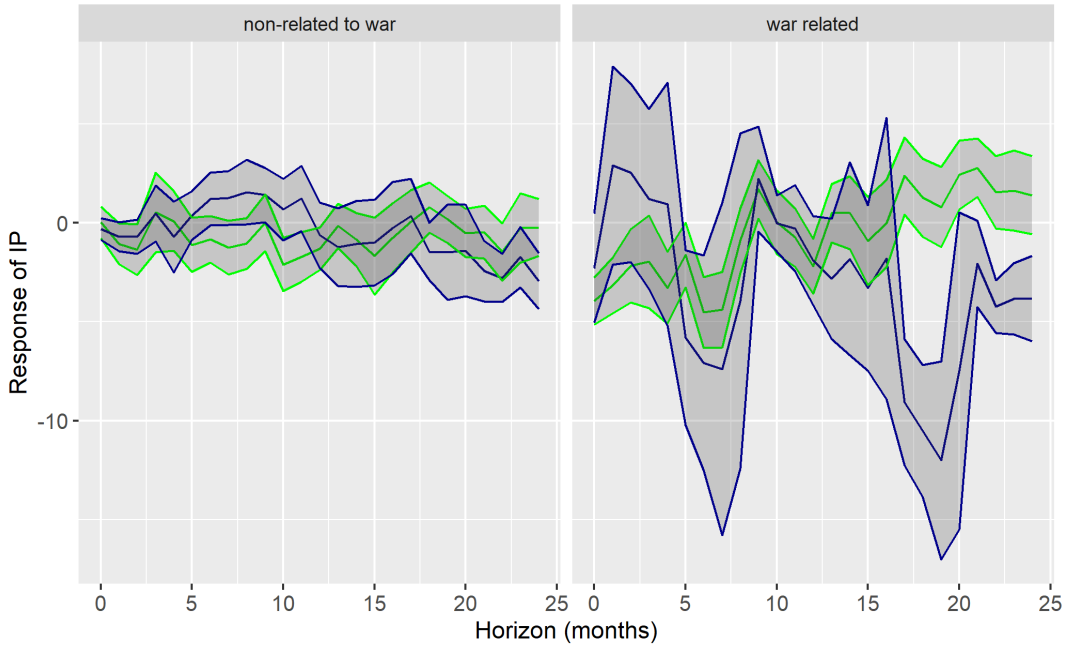


Figure A. 8: sectoral IPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, war-related vs non-war-related



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines - before. N=3922. Source: own elaboration

Figure A. 9: sectoral PPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, high and low government revenue share

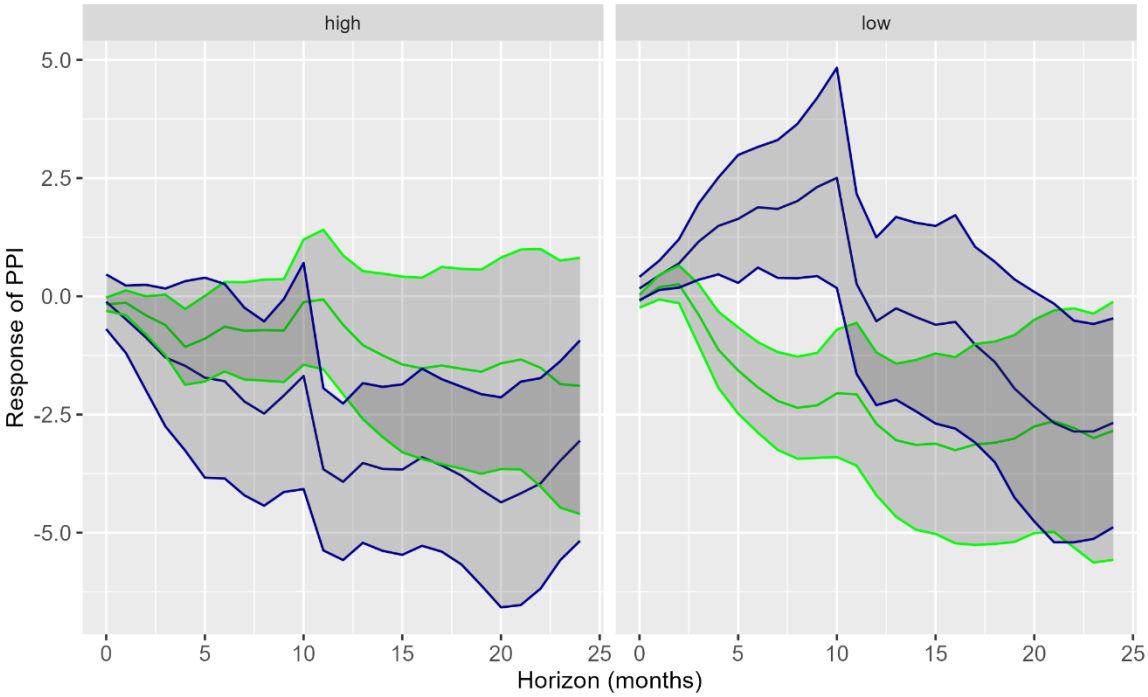
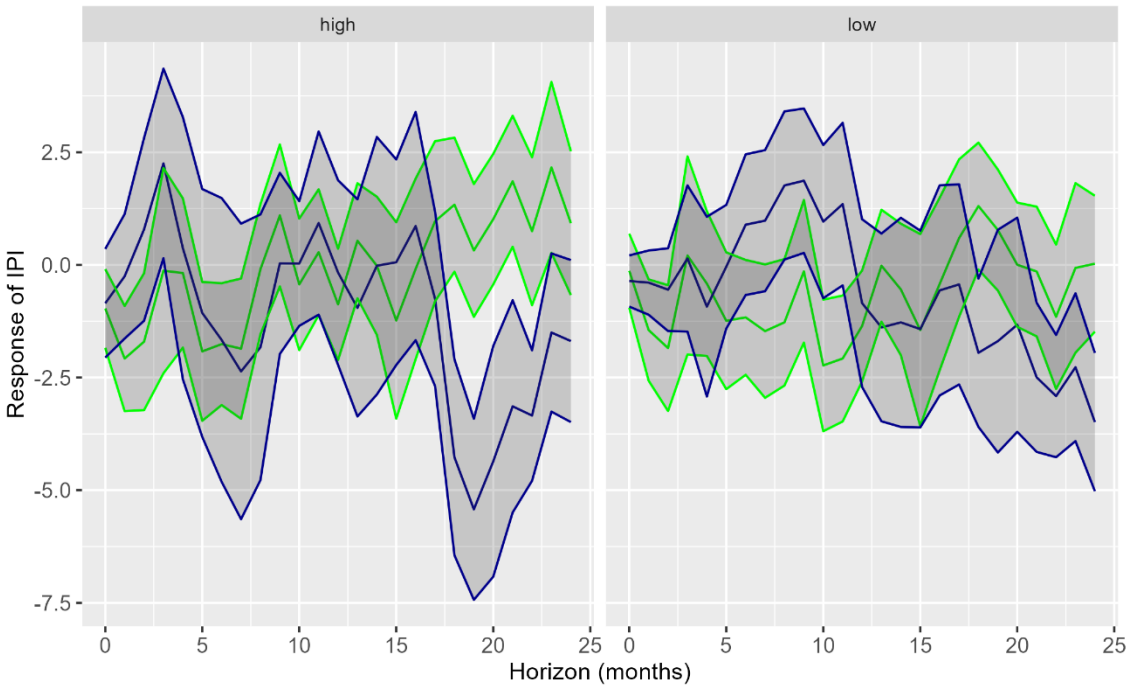


Figure A. 10: sectoral IPI IRFs to a 1 pp shock of the 3-month ROISfix, before and after February 2022, high and low government revenue share



Note: all confidence intervals are at 90%, errors clustered at the time dimension. IRFs were obtained through LP regressions in long-differences with an ample set of controls (with two lags each) described in Section 5. Blue lines represent responses to a monetary policy shock after the start of the war in Ukraine, green lines - before. High revenue share ≥15%; low revenue share <15%. N=3922. Source: own elaboration